Group Project (Assignment 2)

CAB420, Machine Learning

This document sets out the instructions for the CAB420 group assignment (also referred to as Assignment 2). The assignment is worth **30%** of the overall subject grade, and is to be completed in groups of 3 to 4. Each student's contribution should be indicated by a clear explanation of the contributions, and the percentage of the whole work. Please note that your mark may be moderated depending on your individual percentage contribution specified in your report.

Instructions

Students are to select a machine learning problem/topic to investigate in groups of $3 - 4$. You may select your own project idea, or choose from a small list of provided topics. The problem/topic should ideally:

- Have one or more established datasets that can be used.
- Be capable of being addressed in multiple, diverse ways (for example, classification can be done using various formulations of deep learning models, an SVM, a random forest, or several other techniques).

Groups are to implement multiple different methods (one method per group member) to address their chosen problem/topic, and prepare a report that details these approaches and compares their performance on the chosen data. Methods should be appropriate for the problem at hand, and be supported by relevant literature. You are encouraged to discuss possible project ideas with teaching staff during practical sessions or the lecture consultation session.

Students are to self allocate into groups, and are encouraged to use their practical class and/or the teams channel to find a group. Upon forming a group students should:

- Send an email to `cab420query@qut.edu.au`, cc'ing all group members, advising that the group has been formed. You will be advised of your group number by return email.

Submission Details

The project comprises two separate pieces: an optional project proposal, a final report. Note that the first (project proposal) is optional and not assessed, though it is strongly recommended you submit this to receive feedback on your proposed project. Details of these components are outlined below. Note that only one group members needs to submit the project proposal and final report. Please check Canvas for exact due dates.

Project Proposal, Due Week 8

A brief project proposal should be submitted at the end of Week 8. This should be one to two pages in length and should include the following information:

- Team members
- Project title
- Project motivation and objective
- Dataset(s) to be used and brief details on the proposed evaluation protocol

Note that no mark will be recorded for the project proposal, but you will receive feedback on the suitability of your proposed project and approach.

The Report, Due Week 13

The report should be structured as follows:

- Title page, containing project title, team number, and names of team members
- Executive summary, which should be **1** page long and briefly outline the problem being considered, the methods selected, and overall findings.
- Main body of the report, which should be **10–15** pages long and include the following:
  - Introduction/Motivation: clearly motivate your project, and describe the research question, and how it relates to previous works that have been done in this area.
  - Related Work: briefly describe a small number of relevant existing approaches, their respective strengths/weaknesses and relation to each other (i.e. does one build upon another), and the objective of your work.
  - Data: clearly describe the data set, any pre-processing that is performed on the data, any challenges or problems of note in the data, and how the data is split into training, validation and testing sets.
  - Methodology: Clearly explain the three or four algorithms (one per group member) that you used with citations to the literature where appropriate. Please note that ideally your project will extend existing approaches in some manner.

You don't need to propose a novel algorithm, but you might be looking into approaches that have not previously investigated on your dataset. Note also that all considered approaches should be different. For example, rather than simply using three deep convolutional neural networks (DCNN) for a classification task, you could perhaps use (depending on the task) one DCNN, one transformer, and one non-deep learning method such as a random forest (including appropriate feature extraction).

  – Evaluation and Discussion: Present the results of all your approaches clearly, and compare them with existing published results (if possible), and with each other. Discuss why your methods are working better/worse than the existing approaches and each other. The evaluation should also consider the computational demands of the methods, and provide training and evaluation times alongside relevant performance metrics.
  – Conclusions and Future Works: Clearly explain if the experiments match the objectives, the advantages/shortcomings of the proposed approached, and if any changes are required/ plans you have for the future investigations.

- An appendix that details the contribution of each group member towards the project should be included. This should list what each group member contributed, as well as an approximate overall percentage that each student contributed. This statement should be signed by each group member.

You may optionally wish to include further appendices to include items such as code or additional (non-critical) results. Though please be aware that all critical content should be included within the main body of the report, and content within the appendices will be considered supplementary.

Video Presentation, Due Week 13

A short (5 minute) pre-recorded video presentation is due alongside the report. The presentation should:

- Describe the dataset used;
- Briefly present the algorithms investigated;
- Provide a brief discussion of the results achieved.

Depending on your project, a small demonstration of the method and/or displaying sample outputs may also be appropriate.

Groups may record their presentation using software of their choice, but presentations must be encoded in a video format that can be played by VLC (`https://www.videolan.org/vlc/`).

Submission

Materials should be submitted via the provided Canvas links. Only one group member per group should make a submission. If multiple group members submit, we kindly request that you contact `cab420query@qut.edu.au` to advise the teaching and marking team that this has occurred.

Topic Suggestions

For this assignment, you can propose your own project ideas. If you wish to do this, please contact `cab420query@qut.edu.au`, or discuss the idea with the teaching team during practicals, or after lectures to see whether the proposed project is suitable for this assessment task.

You may also wish to look at Kaggle competitions. Kaggle is an onlinecommunity of data scientists and machine learners, owned by Google. One of the more interesting aspects of Kaggle is that it provides datasets and organises many competitions for machine learning tasks. It also manages a leaderboard where the participants can publish their results. You are free to browse the competitions there to see if there are any challenges you are interested in, or if there are any datasets that you think may be of use for your project.

Please note that any dataset that has been used in a CAB420 example or practical is not suitable for this assignment.

When formulating ideas for your project, we advise you to consider the following:

- Is there a good dataset that can you can use? Collecting and/or annotating your own dataset is not advisable as this can be a very onerous task. Using an existing dataset is highly recommended.
- Is the dataset a suitable size for your problem? A dataset that is too small will limit you to using simple methods, while have a dataset that is too large may make it hard to train models for some (or all) group members. Avoid tiny datasets, and be prepared to only use a portion of a large dataset if compute is limited.
- Are your dataset splits defined? Some datasets will come with pre-defined training, validation and testing splits. For others, you will need to do this. Make sure that all group members use the same data splits.
- Are your methods diverse? The methods that you employ should be varied. Simply taking the same pre-processing code and changing an SVM to a Random Forest, or grabbing a different pre-trained CNN backbone, is not ideal. Variation can be introduced in many ways such as:
  - by varying the machine learning methods themselves;
  - by changing the pre-processing used to prepare the data;
  - for deep networks, by varying the broad type of network used (i.e. CNN vs Transformer) and/or the training paradigm chosen (multi-task learning, self-supervised learning, metric learning, etc).
- What pre-processing do you need to do to your data? Does this vary for your different methods? Remember that while deep networks often work well with raw data, for non-deep learning models this is rarely ideal (tabular data, with appropriate encoding of categoricals, is the possible exception here). Carefully consider how data should be prepared, and discuss this with the teaching team if you are uncertain.

- How should your methods be evaluated? You should be using the same metrics and evaluation protocol for all your methods. You are also encouraged to dig more deeply into the results to uncover differences in performance between models. Are there perhaps some types of samples that one model is consistently better (or worse) at than another? Can you identify any reasons as to why this might be?

A small selection of project ideas that you may also choose from follow.

Enron E-mail Classification

The Enron E-mail data set contains about 500,000 e-mails from about 150 users. The data set is available here: `http://www.cs.cmu.edu/~enron/`. Can you classify the text of an e-mail message to decide who sent it?

Object Recognition or Clustering

The Caltech 256 dataset contains images of 256 object categories taken at varying orientations, varying lighting conditions, and with different backgrounds, and is available at `http://www.vision.caltech.edu/Image_Datasets/Caltech256/`. You can try to create an object recognition system which can identify which object category is the best match for a given test image. Another approach may be to apply clustering to learn object categories without supervision, where you could apply clustering to different learned representations (i.e. HOG features, deep network embeddings, etc).

Please note, comparing classification and clustering within the one project is not recommended as these are not easily compared. These are intended as two separate suggestions.

Speaker Recognition

Speaker recognition is the task of recognising someone by the way that they speak. This is a classical problem in biometrics and machine learning. The Common Voice (`https://voice.mozilla.org/en/datasets`) dataset contains a large number of speakers and associated meta-data, and spans multiple languages. You could simply try to recognise a speaker, or investigate the impact of training and evaluating a model on different language, or explore how meta-data could be used to improve performance.

Crowd Counting

Crowd counting is the task of counting the number of people in a scene. The output of crowd counting can be simply a single number representing the total number of people in a scene, or a density map that indicates how people are distributed in a scene (or both). A large number of datasets have been released for crowd counting (see `https://github.com/gjy3035/Awesome-Crowd-Counting#datasets`). Using one or more of these you could investigate crowd counting methods, or explore how different types of methods generalise to different conditions.

Semantic Segmentation of Aerial Data

Semantic segmentation is the task of labelling each pixel of an image with a label indicating what is at the pixel. This is commonly used within scene understanding pipelines to identify objects and regions of interest in a scene. DroneDeploy have released a segmentation dataset and benchmark suite (see `https://github.com/dronedeploy/dd-ml-segmentation-benchmark`) for semantic segmentation from drone data. Along side colour imagery, elevation data has also been captured. How could you combine elevation and RGB data to improve segmentation performance?

Other Sources of Project Ideas

There are many other places you may turn to for project ideas. A few useful links include:

- `http://cs229.stanford.edu/projects2015.html`
- `http://cs229.stanford.edu/projects2016.html`
- `https://github.com/NirantK/awesome-project-ideas`